Sizhe Li

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EDUCATION

The Chinese University of Hong Kong, Shenzhen *Shenzhen, China*
Bachelor of Science in Statistics *Sep. 2022 – Present*
Cumulative GPA: 3.923/4.000 (rank: 6/329),   Major GPA: 3.982/4.000

University of California, Berkeley *California, USA*
Visiting Student *Aug. 2024 – Dec. 2024*
GPA: 4.000/4.000

No.1 Middle School Affiliated to Central China Normal University *Wuhan, China*
High School Diploma *Sep. 2019 – Jun. 2022*

RESEARCH EXPERIENCE

Prediction-Specific Design of Learning-Augmented Algorithms *Feb 2024 – May 2025*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- **New Framework & Definitions:** Introduced a *prediction-specific framework* for learning-augmented algorithms enabling fine-grained, per-prediction performance guarantees. Proposed a novel definition of *strong optimality* capturing Pareto optimality in both consistency and robustness tradeoffs. Proved the suboptimality of state-of-the-art methods (NeurIPS 2018, 2021) in this new framework.
- **Generic Approach:** Developed a *bi-level optimization* framework for designing strongly-optimal algorithms, applicable to diverse online problems.
- **Explicit Algorithm Design:** Designed provably strongly optimal algorithms for *deterministic ski rental*, *randomized ski rental* and *one-max search*.

Optimal Tradeoff in Learning-Augmented Non-Clairvoyant Scheduling *June 2025 – Present*
Supervised by Prof. Tongxin Li and Prof. Nicolas Christianson

- This work advances a central open problem in learning-augmented online scheduling: determining the optimal consistency–robustness trade-off for general n jobs. Prior bounds (Wei & Zhang, NeurIPS'20) were tight only for $n = 2$ and two extreme points. We prove that the classical lower bound is not tight for $n \geq 5$, establish refined lower and upper bounds for finite n along with the matching algorithm, and derive analytical lower and upper bounds in the asymptotic regime as $n \to \infty$, providing a near-optimal characterization of the frontier. A first-author preprint will be released soon.

PUBLICATIONS

- **Sizhe Li**, Nicolas Christianson, Tongxin Li. *Prediction-Specific Design of Learning-Augmented Algorithms.* arXiv:2510.14887, arXiv preprint. **ACM SIGMETRICS 2026**.

PROJECTS

Simulation Model for ICU Resource Optimization, UC Berkeley *Aug. 2024 – Dec. 2024*

- Developed a discrete-event simulation to optimize ICU capacity using the MIMIC-IV demo dataset.
- Proposed patient prioritization based on severity and urgency, using dynamic penalty evaluation.
- Designed and compared three resource allocation solvers under varied load conditions.
- Conducted sensitivity analysis to evaluate model robustness and key performance drivers.

TEACHING

Undergraduate Teaching Fellow, CUHK-Shenzhen

- **MAT2041: Linear Algebra and Applications** Fall 2023, Spring 2025
- **CSC1001: Introduction to Computer Science: Programming Methodology** Fall 2025
- **DDA4001: Stochastic Processes** Spring 2026

HONORS AND AWARDS

- Dean's List, 2023–2025
- Academic Performance Scholarship (Top 1–2%), 2023–2024
- Undergraduate Research Award, 2025

SKILLS

- **Algorithms and Theory:** Approximation algorithms, competitive analysis, complexity theory
- **Programming:** Python (NumPy, Pandas, SciPy, Matplotlib), Java, R, MATLAB
- **Tools:** Git, Linux, Jupyter, VS Code, Overleaf
- **Typesetting:** LaTeX